Plexus Discovery

Progress note — written to be read, not decoded

31 July 2026

Plain English throughout. Organised by phase, because phases are where we stop and you decide.

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1. What we are doing — two things at once

Track A — build the method. An automated loop that searches for biological *mechanisms* rather than parameter values. It proposes a change to the model's structure ("remove the pushing force", "add oriented cell division"), writes down what it expects *before* running, runs the simulation, measures, and records whether it was right. The discipline that makes it worth building: a change of numbers can never masquerade as a new idea.

Track B — reproduce Okuda's result, as the proof that Track A works. Okuda et al. (2018) grew a hollow ball of cells that spontaneously forms tubes, undulations and branches, driven by a chemical pattern on its own surface. We are rebuilding that here. If the loop finds the mechanism largely by itself, the method is demonstrated. *The figure is not a side quest — it is the evidence.*

The two tracks feed each other: the reproduction is the honest test of the method, and the method is what makes the reproduction more than hand-tuning.

2. What we believe about the biology

Nothing *measured*. But we now write down what we assume.

The register of *results* is still empty, and should stay empty until a batch has been taken with instruments that work. Every number we have was measured with at least one broken ruler in the path.

What changed on 31 July is a different thing: the register of **assumptions** is no longer empty. Eleven basics about cell tissue are written down in `discovery_okuda/PREMISES.md` — not a literature review, but the minimum a person needs in order to look at one of our runs and say "that cannot be right". Each is phrased so that a computer can check it against a run, and each one now does (Phase 1).

The distinction matters. A result is something we found out. A premise is something we are *assuming*, stated plainly enough that it can be caught being wrong. Writing the second kind down is not a claim about biology — it is a way of making our own mistakes visible.

3. Two questions instead of one number

Success is not a threshold a single run clears. It is two questions, and they are asked separately because they fail separately.

1. What can we reproduce?	Each of the paper's morphologies is a row, scored qualitatively: *not attempted / attempted and failed / partial / reproduced.* **Today 0 of 4.** Budding is tracked too — our substrate produces it and no criterion ever mentioned it.
2. What do we know?	Already measured, and previously subordinate to the tube gate: map coverage **27%** (3 of 11 cells can state a verdict) — and all of it is routing; solo operator effects are **0 of 8**. Nine claims resolved, ten open, surprise rate **0.375**.

4. The team of agents

`ROLES.md` is the source of truth, and the code is checked against it in both directions by `roles.py --check` — the same discipline `PREMISES.md` has for the biology. The roster had reached sixteen roles nobody could account for, because it lived in four places at once: the code, a table in this note, a figure drawn by hand, and everybody's memory. Four descriptions of one thing are three descriptions that are wrong. There is now one, and this note does not repeat it.

The roster was judged by a ledger, not by argument. A live round spent 23.5 minutes across sixteen calls, of which the Proposer took 14.8 — 63% of all agent time — while the Judge and the Referee were *never called at all.* That is not a claim that they were redundant; it is a record showing they had nothing to do. The cause was a borrowed design: *Co-Scientist* runs a tournament of hypotheses because it performs no experiments, so its proposals can only be debated. We measure. Where a tournament would rank two ideas by argument, we run both.

Three acts, and the division that matters. Act 1 proposes, Act 2 measures, Act 3 decides. Within that, the load-bearing split is between what is *arithmetic* and what is *judgement*: the Critic decides whether a batch **can** be run and its refusals are enumerable and arguable-with-never; peer-review decides whether it is **worth** running, which is a judgement, and it advises rather than vetoes. A judgement that can silently cost twelve GPU-hours is a veto wearing an adviser's name.

A round that produces no evidence is aborted and routes back to Act 1 carrying the refusals. It is recorded in full but does not increment the round counter and enters no coverage denominator — counting it as a normal round is exactly how a log came to tell the Proposer `coverage 0%`, from which it correctly concluded its own ledger was broken. Two consecutive aborts stop the campaign and ask, because a second abort means the refusal reasons were not actionable, which is a fact about us rather than about the search.

As of Phase 8 the loop can be run with no agent in it at all (§8.1), and every hand-off between these roles is declared in `WIRING.md` and checked (§8.2). Those two are what make the paragraphs above testable rather than remembered.

5. The phases

Eight stages. **I stop at the end of each one and wait for you.** That is the partition.

Phase 0 — Pre-flight: fix the instruments done (28 of 28)

Goal. Make every measurement trustworthy. Nothing runs unattended until this lands. Not one of the twenty-eight items is science; each one makes a question *askable*.

The last two, closed. *Comparing a batch at a common frame* — the per-run evidence horizon was right and did not make a BATCH comparable. If one run tears at frame 300 and another holds to 700, comparing their "final" states puts a torn mesh beside a healthy one and reports the difference as biology: both operands valid, the comparison wrong, which is the shape of every error that has cost this campaign a batch. A batch is now read at one frame, the earliest horizon in it, and the tool reports what that costs — how much of each run is discarded, and a warning when one run's early death is being imposed on everyone. *The explicit vocabulary* — `elongation`, `elongation_peak`, `elongation_at_end`, `elongation_unforced`. Writers emit both names, readers resolve either, the archive is never rewritten: cutting 92 references over in one step leaves a window where a key is written by half the code and read by the other half, and since an absent metric reads as "not measured", the bug would surface as a round that quietly learns nothing.

The repairs worth knowing about

what	what was found
survival test	It relaxed a fresh sphere and returned 1.014 for every run. Now real: **1.415 → 1.325, so 94% of the elongation survived the drivers going off** — the first honest survival number the campaign has produced.
the camera	Growth was drawn as *shrinkage*: apparent size fell to 0.52× on a run that grew by half. A second viewpoint justified itself immediately — in one run **the tube is invisible from the side and obvious from the top**.
the tube bar	**No capsule of any aspect ratio can reach the old bar of 2.0** (ceiling 1.90). The criterion was not asking for a tube; it was asking for the spike artefact.
damage	The blended damage score was counting *cell division*: sliver correlates with tip count at +0.94, while genuinely broken cells stayed at **0 for 300 frames**.
the 1778 ceiling	Arithmetic, not physics: (3552+4)/2 = 1778, and all 32 runs of the overnight study stopped there.
the missing arrow	The emitter never passed the implementation through — the ablation would have reported "no effect" *without ever running the new code.*
the curve plot	Drawn every run since the beginning, referenced **nowhere in the codebase**.
timing	The round reported `calls: 0` for a round that made about 25.

Phase 1 — Teach the loop what a tissue is: the premise gate done

Why this became its own phase. Ten defects were found in one day, every one by Cedric looking at a picture, and all of the same kind: an operator or an instrument not doing what its name says. The loop checked that runs finished, that the mesh was valid, that the numbers were finite. It never once checked that the thing it had simulated was a tissue.

Eleven basics, each one something a computer can run and fail, in three tiers by cost: read the recipe (instant, refuses a hopeless run before a GPU is touched); read the run (free, goes loudly into the record); and run a small experiment of its own — switch everything off and let the ball sit for forty frames; it must not change size.

A broken premise must be declared, not silenced. Switching off growth to ask what a protrusion is made of genuinely breaks premise 1, and that is science. So a recipe may waive a premise *in writing, with a reason*; an undeclared violation is a hard stop. That is the only thing separating "we tested an idea" from "the settings were wrong".

Every check was proved to fail before being trusted. The one to look at: run against the code as it stood that morning, a resting ball collapses **5.00 → 0.69** in forty frames. That single check, which costs about a minute, would have caught the worst defect of the campaign on day one.

Phase 2 — Make Okuda's settings reachable done (6 of 6)

The box used to describe the work without listing it, so the count had no visible denominator. All six, with what each one turned out to be:

. item	state	what it turned out to be
0. The ball could not get big enough	done	*Not on the list.* A spring held the shell at the size it was first built at while the cells inside grew sixteen-fold, so the sheet folded through itself. The spring's target now follows what the cells are asking for: the same run grows smoothly to 5 240 cells and stays clean.
1. Widen the out-of-range settings	done	Done, and the new bounds are *derived* rather than chosen: growth sharpness to 49.5 (Okuda needs 10), inhibitor spread to 12.5 (he needs 10), reaction speed unbounded.
2. Replace the mislabelled length-scale	done	The quantity we called a pattern size is a solver rate. Resolved in Phase 4 by measuring the pattern instead of setting it.
3. The shape-to-chemistry feedback	done	One contract, four implementations — curvature, tension, apical area, pressure — because *which* feature the chemistry listens to is the hypothesis, not a dial. Force and size were refused with reasons.
4. A fixture asserting his published constants	done	His Table 1 is now written down in the Grounder, quoted, with the sentence each value comes from — the Hill coefficient, the two diffusivities, the cell cycle, and the two regime *inequalities* that actually define his setting. Asserting a run against it is Phase 3, because it needs the Grounder wired.
5. His cell counts, with the buffer sized for the *destination*	done	**The item that voided the weekend batch**, and it is measured rather than argued — see below.

Phase 3 — Switch on the agents that never run done (3 of 3)

Ablation is compulsory. A causal claim needs two experiments: adding a mechanism and seeing the shape appear shows it is *sufficient*; taking it away and seeing the shape die shows it is *necessary*. Either alone is an opinion. Every claim now declares its kind and the checker throws out a batch that asks for more than it tests, *before* any simulation starts. Sufficient earns "is sufficient for"; both earn "causes"; neither earns "is associated with". **The word can no longer outrun the experiment.**

The Grounder is wired. The only agent that reads the paper had no call site anywhere — every entry point was exercised by its own smoke test alone. Batch construction now asks it, and it answers with Okuda's numbers carrying the sentence they came from. They arrive as *parameters*, so composition identity is unchanged: setting the starting size is a retune, not a new mechanism. It *advises* — the faithful share of a batch inherits his numbers, the exploratory share is free to leave them, because a search that may only stand where the paper stands cannot discover that the paper is not the only place to stand.

Phase 4 — Measurements that can tell his regimes apart

done

The classifier. Sphere, undulation, tube, branched, plus **invalid** (the surface passes through itself, so no shape reading means anything) and **unclear**, returned deliberately near a boundary. A classifier that always answers can never be wrong, and one that can never be wrong teaches nothing. “Invalid” beats every morphology — this campaign has already once reported a crumpled shell as a result. Until 1 August nothing called it: built, certified, never asked.

The calibration, and the surprise. The plan was to tune until Okuda’s handful of spots appeared on a 2000-cell ball, then freeze. Counting spots over time — for the first time — showed the pattern never settles: 104 spots at step 0, 19 at 100, 7 at 200, and **1 by step 400**, where it stays. It keeps merging. So “the right number of spots” is not a setting to be found but **a moment to be chosen**, and the choice has to be stated rather than tuned into.

Phase 4b — Can the chemistry be simulated at all?

done

Why this became a phase mid-flight. The pre-launch stopped on a run whose chemistry turned to infinities within a hundred frames. The rule that exists to prevent exactly this had been **fifty times too generous** for as long as the campaign has existed.

How. The stability condition was written down as a *comment*, worked out once, for the settings of the day. Then the defaults changed, and Phase 2 deliberately widened the ranges so Okuda’s own values would be reachable — and the arithmetic in the comment quietly stopped describing the code. On top of that it counted one factor twice, reporting a fiftieth of the true figure. **A comment cannot be re-derived; it can only be re-read, and nobody had cause to re-read it.** The condition is now computed from the settings actually in force, and a composition that breaches it is refused before it reaches a GPU rather than after.

Phase 4c — Multi-spot initialisation, and replication

done

Why. Cedric: *“did you forget to run multiple tubes from now on, not a single one”*. He is right, and the note already said so — one protrusion per run is a single sample, and you cannot tell “this mechanism makes tubes” from “this run made a tube”. Several spots give within-run replication for free, and it is what Okuda shows. Every control I built during the pre-launch used one spot or none.

Measured, six runs at the stable point, 300 frames:

seed fraction	act_max	activated fraction	
0.02	0.423	0.40	
0.06	0.501	0.37	three seeds, **identical to 3 d.p.**
0.15	0.598	0.12	
0.30	0.000	0.10	the pattern is extinguished (P4)

Two findings, one of them a problem. Seeding more widely does not give more pattern — it gives *less*, and past 0.15 it gives none at all: a seed covering a third of the shell leaves no room for the inhibitor to separate domains, so the whole field decays together. The useful band is narrow and sits below 0.15.

And the three seeds were bit-identical — 0.501, 0.501, 0.501. `general.seed` was written into every spec and read by *nothing*: both stochastic operators take a `seed` (the mesh jitter, and which cells are nucleated) and the translator passed neither. “Three independent runs” were one run repeated, so every spread this campaign has quoted across seeds described the floating-point reproducibility of a single trajectory.

Fixed, and the fix follows the working specs rather than my first idea. I offset the two operators' seeds, reasoning that jitter and nucleation are independent draws. True, and beside the point: `archive_rounded.py`, `archive_vh_rd_coral.py` and `run_tyssue_fig5.py` all pass *one shared* SEED to both, and a composition that reproduces an archived run has to draw the numbers that run drew. Departing from the convention would have made every reproduction check compare two different experiments. A spec that declares a seed no operator consumes is now refused outright.

Replication, measured for the first time — three seeds, 300 frames, same composition:

	act_max	**activated fraction**	**elongation**
before	0.501, 0.501, 0.501	0.374 ×3	1.006 ×3
after	0.501, 0.447, 0.483	0.407, 0.379, 0.374	1.006, 1.006, 1.005
	sd **0.022**	sd **0.015**	sd 0.0005

That last row is what a difference between two compositions now has to beat to mean anything. It did not exist yesterday, and every claim made without it was comparing single samples.

Phase 5 — The first live run: the agents do not talk to each other DONE

It ran, and that is the point. Two rounds, eight simulations, 23 agent calls — and **one usable run, map coverage 0%**. The machinery worked; the science did not, and the reason was the same everywhere.

Not one finding reached another agent. The Critic refused eleven slots and the Proposer was never told why. The Biologist judged every specimen and spoke to nobody. An audit of every hand-off found the same shape **seven more times**: a role produced a correct judgement, wrote it down, and no one read it. *The campaign's defining failure was not bad judgement — it was correct judgement with no recipient.* Drawing the graph is what made it obvious.

Two consequences that are now permanent. Refusals became evidence: a slot the Critic rejects is written to `batch_refusals.jsonl` and lands in the next Proposer's prompt, because a loop told only about its successes cannot avoid repeating its failures. And the Biologist reads `PREMISES.md` mechanically, so the premise gate is a document a person can argue with rather than a rule buried in code.

The black movie, which is the phase in miniature: frame 0 was perfectly good and was drawn black because frame 800 diverged — the colour scale was normalised over the whole run, so one bad frame erased eight hundred readable ones. Nothing was wrong with the simulation or the plotter alone; they disagreed about what the scale was for.

Closed by Phase 8, which is what this phase was asking for without being able to name it. `WIRING.md` declares every hand-off and `wiring.py` fails the round when an artifact is written and never read — the seven-times-repeated shape becomes a failing check rather than an audit somebody has to remember to run.

Phase 6 — The rebuilt loop, and its first launch DONE

Goal. Rebuild the roster around measurement rather than debate, and launch it.

What the cold start found, and what it cost to misread it

Across all 63 finished runs, `n_protruding` was zero everywhere. **This project has never produced a protrusion.** The loop reported that correctly — and then gave up, which was the expensive mistake, and it was ours. **Track A asks a different question**: a composition that patterns without protruding is a *point on the map*, not a failure. On the corrected ranking, **46 of 63 runs serve Track A**. Only Track B can run out of things worth running; the operator landscape always has more of itself to understand.

Closing Phase 6 — what was run

Nine rounds, 2 August 2026. **79 hypotheses posed, 58 admitted as evidence, 21 refused.** Ten defects found and fixed while it ran — among them: the reading loop recorded *one* run per round and that run was a chimera, assembled from different runs' fields; and the ranking read `premises_broken` from the wrong dictionary, so it never once saw a broken premise.

Results, and what we conclude

The machinery works. 30 predictions were refuted and 23 confirmed — a surprise rate that means the predictions were real rather than safe. **Tubes appeared, on sound specimens. But the ceiling was set in round 3 and never moved.** No valid specimen exceeded `protr_peak` 1.295 or two tubes; every run above two tubes was invalid — dead chemistry or a self-intersecting sheet. And the valid frontier contains **no reaction–diffusion at all**: `cell_rd_seed` and `morphogen_growth_3d`, but neither `cell_diffuse` nor `cell_react`. The shape the campaign can make is not yet the shape the paper is about.

The main conclusion: the logic did not emerge

It was not in the agentic loop, and it was never going to arrive by itself. Nine rounds of capable agents produced conclusions no evidence supported — "the ceiling is FUNDAMENTAL", "division is NECESSARY" — because nothing in the loop could tell an untested possibility from a refuted one, and *could be* had nowhere to live. A campaign that cannot represent "not yet tried" will write "cannot be" instead. That is what Phase 7 exists to fix, and it is a *structural* fix: the unearned conclusion has to be unwritable, not discouraged.

Phase 7 — The agentic loop update: giving the loop a logic DONE

Goal. Supply the reasoning discipline Phase 6 proved will not emerge on its own — as *structure and code*, not as another voice asking the others to be careful. No new agent.

Why one phase and not two. *Absence of evidence read as evidence of absence* and *absence of an instrument read as absence of an effect* are the same sentence with a different subject. Both are cured by the same move: make the unearned conclusion **unwritable**, and give the loop somewhere to put what it has not yet tried.

`LOGIC.md`, and the checker that makes it real. A third source of truth beside the two that already hold — `ROLES.md` kept honest by `roles.py --check`, `PREMISES.md` by the Biologist. **`logic.py --check` refuses a malformed claim at the moment the Meta-review writes it**, so a bad claim never enters the document the Proposer reads. A rule agents are merely *asked* to honour is the failure mode being removed.

Four modalities, and a home for *could be* — which was previously unrepresentable, so a campaign that could not say "not yet tried" wrote "cannot be" instead.

can be	demonstrated positive	one instance suffices
cannot be	universal negative	must carry its quantifier
cannot not be	necessity	the most expensive claim of all
could be	untested, open	**free — and today unrepresentable**

Independence is a distinct condition-signature, not a run count. Ten GM-with-division runs collapse to *one* observation. A negative needs three; **a necessity claim needs three and a failed rescue attempt** — a different experiment, not a re-reading of the same one.

The ablation asymmetry, which is Cedric's and is the sharpest idea in the phase: the same experiment supports a positive strongly and a negative weakly. A null ablation must repeat on an independent background; a positive passes on one.

No conclusion about an unmeasured property. `morphology=sphere` was recorded for the run carrying the finest Turing pattern in the campaign, because the shape was measured and the pattern was not. The honest record is *not measured*, and the honest next step is a request for an instrument — which the Metrologist now builds **on demand rather than on schedule**, so it costs nothing in the rounds that need nothing.

Closing Phase 7

The phase turned out to be wiring, not construction. Almost everything it needed already existed and had never been called: `CLAIM_KINDS`, the batch and memory checks, the reservoir counters, the pattern-scale instrument — already *certified* — the status setter, the allowed-verb table. The external audit's closing line is the phase in one sentence: *"writing it to a file is not the same as handing it over."*

The false start of 3 August. The ceiling cried wolf; the clean start was not clean, and handed a fresh Proposer a memory saying `THIS BODY IS CLOSED`. The lesson is the phase's own turned on itself — a rule written but not wired changes nothing, and that applies to the rules *about* rules. Which is why Phase 8 begins by making the loop runnable with no agent in it at all: the only way to find a broken hand-off without paying for a round.

Phase 8 — Entities, edges and sweeps: the loop stops being a script DONE

Goal. Three things the 3 August runs showed cannot be patched one at a time: give the loop a *declared structure* it can be checked against, close the audit findings that have been open since the graph was drawn, and schedule the parameter sweep that has been implemented and never run.

Why now, and why one phase

Six defects were fixed on 3 August and *not one of them was a logic error*:

`composition.json`	written by `write_config`, never by recon: a producer with no consumer
`run[:14]`	a display truncation read back as an identifier
`batch_ok=None`	a verdict computed, never applied
menu preamble	a contract asserted in prose, enforced nowhere
`max_per_parent=6`	a limit set for one purpose binding another
`say()` vs the wrapper	two owners of one property, folding at 100 and at 96

Every one is an **interface defect**: the components are individually correct and disagree at the seam. That is the class of bug patching cannot retire, because each patch closes one seam and the next round exposes the next. Sixteen more declared edges are unchecked today.

1. Settled by construction: the loop runs with no agent in it

Every defect above was found by *launching*, which is the most expensive instrument available: a round costs ten agent-minutes and half an hour of cluster before it will tell you that a display name was read as an identifier. **The loop must be testable without an agent and without a cluster.** One seam — `run_agent` — carries every model call, and one — `cluster` — carries every job, so both can be substituted for deterministic stubs that return schema-valid answers, including the *wrong* answers we have actually seen: an edit that names a phenotype, a verdict with no flag, a caption with no JSON, a proposal under the wrong key.

A full three-act round then runs offline, in seconds, for nothing, and the questions it answers are exactly the ones that have cost us rounds: does every declared artifact get written, does every gate fire, does the batch fill, is the record complete before its readers run. A regression becomes a failing test rather than a wasted evening. **This is built first, because it is what makes the rest of the phase checkable.**

8.1 — Built, and what it found immediately **DONE**

Three seams, not two. `run_claude` (not `run_agent` — `agents/` is on `sys.path`, so `llm` and `agents.llm` are two module objects for one file, and patching either alone leaves the other live: the first "offline" round spent \$0.37 while printing that it was free), the cluster, and — found only by running — **the VLM**. A 23 GB model load is not offline by any reading, and a harness that takes five minutes will not be run before a launch, which is the only thing it is for.

`offline.py` — a full three-act round in **≈6 seconds**, no agent, no GPU, no VLM. `test_offline.py` — **8 cases, 8 passing**, each one a round we have already paid for: the prompt arrives whole; no path reaches the real CLI (proved with a `subprocess.Popen` tripwire); a clean round fills its batch and reaches Act 3; phenotype edits are refused and repaired *in the same round*; a proposal under the wrong key is still read; a reply with no JSON does not fabricate a batch; `REJECT` with no flag is still a rejection; a truncated name resolves to its run.

What it found on its first honest run — the worst defect of the campaign. `run_agent` contained `prompt.split("BREVITY")[0]`, and the Proposer's template pastes `{BREVITY}` at character 331 of a 16,093-character prompt. **The agent received 331 characters**: no legal-move menu, no refusals, no history, no output schema. It invented `add branching` and `add chemotaxis` because it had never been shown the operators — and when it reported *"I lack the injected LEGAL MOVES menu (this call omitted it)"*, that was **literally true** and was read here as a hallucination. Four rounds were spent on the consequences, including the recon Proposer naming zero runs, whose JSON schema sat past character 331 as well. Excising the *block* rather than truncating at it: 331 → 18,025 characters, 36 moves, schema present.

Two faults were the harness's own — worth recording, because inventing failures is the characteristic way a fake goes wrong. Inferring the batch size with a loose regex matched "1 slot" inside a pasted refusal block, so it proposed a batch of one; and a hand-written metrics series omitted `act_mean`, so the Diagnostician read all six runs as diverged and stopped the round over an apparatus fault that did not exist. Both now *template from real files* instead of being written by hand, which is the rule this box exists to state: **a fake that invents failures trains you to ignore the alarm.**

Also landed with it. Same-round repair when the Critic guts a batch, because its refusals are mechanical and have a right answer — measured 1 of 12 → 12 of 12 on round 2's real proposal, one extra call, bounded at one so it cannot loop. Attrition written as a number (`asked/proposed/refused/delivered`) rather than two prints a reader must subtract. `REVISE` reads as a rejection. `[trace]` off by default. And "not checkable" no longer fires on recon slots, which carry no prediction *by design* — it was describing the design as a defect, three lines per run, twelve runs a round.

2. The entities, as classes and a registry

`ROLES.md` already declares who exists and `roles.py -check` keeps it honest — but it checks *existence only*. Each role also declares **Sends to:**, and nothing has ever verified that those fifteen edges correspond to a read anywhere in the code. So a `Role` and an `Artifact` become objects in a registry, each artifact naming its writer, its readers and its schema, and the checker fails the round when:

- an artifact is written and never read — five exist today: `causal_descriptions.md`, `diagnoses.jsonl`, `archivist.jsonl`, `holes.jsonl`, `logic_report.jsonl`

- a declared **Sends to:** edge has no corresponding read
- a prompt asserts a guarantee no checker enforces
- a rendering is assigned to an identifier field

This is the same idiom a fourth time — a document that is the source of truth beside a checker that enforces it — applied to the one thing currently specified only in comments.

8.2 — Built DONE

`WIRING.md` declares every artifact, its writer and its readers; `wiring.py` derives the same graph from the code and reports the difference. **The rule that carries the weight needs no analysis at all: a new artifact must be declared, *with a reader.*** It is deterministic, it is the rule that was broken five times, and it caught `batch_attrition.jsonl` — which I had added an hour earlier and nothing read — within the hour.

The derivation is a corroborating heuristic, and says so. Writes go through class constructors (`BudgetLedger(path=...)`, `Register(path=REGISTER)`) and following those statically needs real dataflow analysis. Two rounds of sharpening — function-scoped windows, then module-constant aliases — took the false alarms from about twenty-five to none, because **a checker that cries wolf is one nobody runs before a launch**. Where analysis cannot reach, the declaration decides; that is exactly how `PREMISES.md` works.

The suite is now 11 cases, 11 passing, including one that exists only to make the checker prove it bites: a row declared with no reader *must* be objected to, or a green tick is indistinguishable from a parser that matched no rows at all.

3. The audit findings, closed

The round record written after its own consumers; the Archivist's `roll_back` truncating at eleven characters against a twelve-character name; the Eye-check defaulting `supports=True`, so an unparsed caption reads as agreement; recon runs that cannot seed the frontier because a verbatim copy carries no composition; and `_dbg_shrink.py`, which does not import.

8.3 — Closed, and one that will not close DONE

The eye no longer agrees by default. `j.get("supports", True)` made an unparsed reply — a timeout, a caption the model never produced — read as the Eye-check *agreeing* with the numbers. That is the one direction this role must never fail in: it is the only channel in the loop that looks at *shape* rather than at number, so a silent yes removes the only thing that could contradict the arithmetic, and removes it invisibly. Three states now, and an unheard eye neither supports nor blocks.

The Archivist's identifier is no longer truncated. `comp[:11]` fitted the column and *measurably* collides on the current log — hashes run 12 to 28 characters — and the Archivist names a branch back to the Supervisor *from that table.* A collision there rolls the campaign back to the wrong composition. The region is display and may be cut; the hash is identity and may not. Same disease as `run[:14]`.

Recon runs can be rebuilt into parents — operators, implementations and parameters recovered from the spec, honestly partial about the connections a spec does not carry.

And the finding underneath it, which is not closed. The rebuilt parents are still refused by the Critic: `cell_diffuse0.d_h=2.0` against a declared ceiling of `0.346`, parameters an emitter never receives, `impl: default` for an operator that has no such implementation. **The runs this project has actually made sit outside the space it is searching.** `from_preset` warns of precisely this in its own docstring — *"if a recipe we already trust cannot be built from legal one-edit moves, the campaign is searching a space that does not contain our own evidence"* — and that validation gate is failing. Whether the ranges are wrong or the runs are is **a decision, not a patch**; the loop now says so once per run and falls back rather than pretending.

The harness isolates each round. Sharing the campaign's accumulating state made every successive offline round harder to satisfy — the fourth in a row had every edit refused as a duplicate of its own predecessors. *A test whose result depends on how many times it has been run is not a test.* The real campaign is copied aside and restored at exit.

4. The sweep that was built and never scheduled

`-mode theta` exists, and its discipline is already mechanical: `comp_hash` **excludes** theta, so a retune cannot register as a new mechanism — a change of numbers cannot masquerade as a new idea because the identity function says so, not because an agent was asked to be careful. What is missing is only wiring: `plan()` emits `recon` and `composition` and never `theta`; nothing chooses the base composition, the parameter or the values; and `build_theta_batch` sweeps *one* parameter of *one* composition, so mixtures of parameters are not expressible at all. Track A is not only "does this mechanism matter" but "what does it do as you turn it up", and the second question has never been asked.

8.4 — A parameter change is a move on the menu DONE

Cedric's design, and better than the separate mode it replaces: `set_param` becomes an edit kind beside `add_op`, `remove_op` and `set_impl`, so the Proposer chooses structural and parameter moves from *one* list and can **mix them in a single batch** — instead of a whole round having to be one or the other.

The discipline holds by construction, not by instruction. `comp_hash` already excludes parameters, so a retune yields the *same* composition identity and can never register as a new mechanism. "A change of numbers can never masquerade as a new idea" is true because the identity function says so — asserted as a test, not asked of an agent.

Two points per parameter, taken from each operator's own `(lo, hi, default)`, so a sweep cannot leave the space the Critic admits. **The specimen is not an axis**: substrate parameters, `n_cells` and `seed` are excluded, because `round.py` grounds starting conditions for a reason it learned when two slots ran 500 cells against a control at 2000 — any difference between them confounded the edit with a fourfold change in specimen size.

Structure first, numbers rationed. Parameter moves outnumbered structural ones seventeen to seven on the current frontier, so a menu of any length would have shown mostly dials and a batch would have gone on turning them. Every structural move is offered; parameter moves are capped at a quarter. Track A asks whether a mechanism *matters* before it asks what it does as you turn it up. The preamble now also states what an operator absent from the menu means — *it does not exist in this system, however real the biology* — which is exactly what round 2 needed and was never told.

What you get. A loop whose parts are declared and checked rather than remembered, with the sweep arm switched on. *Nothing relaunches until the checker passes* — and the checker is now `python test_offline.py`: **12 cases, 12 passing, no agent and no GPU**.

Phase 9 — The demonstration NOT STARTED

Goal. Run the designed grid — six to eight simulations — that produces the figure: tube, undulation and branching, at Okuda's settings.
What you get. *The figure.*

Phase 10 — The method claim NOT STARTED

Goal. Write up what the loop found by itself versus what we told it, with the record of predictions and retractions as the evidence.
What you get. The argument that Track A works, resting on Track B.

6. The 31 July lesson, in three lines

Ten defects in one day, every one found by Cedric looking at a picture and none by the loop — a ball that shrank to nothing, a movie where every cell turned green at once, a coral that began uniformly red. They were all the same kind: the loop could tell whether a *simulation* succeeded and had never been able to tell whether the *specimen* was a tissue. Phase 1 is the answer to exactly that, and the only way to find out whether it is enough is to run a batch and see whether the gate catches anything before Cedric does.

7. What I need from you

Nothing blocking. Two things worth knowing:

1. **The buffer gate is the one decision worth taking now.** Phase 5 has already been spent once on runs that could not have produced evidence. A pre-launch refusal costs nothing and would have saved the batch.
2. **Tell me if this note is the right shape.** Too long, too short, wrong order — it is cheap to change, and it is the thing that lets you steer.

Appendix — words I could not avoid

Only these four. Everything else above is ordinary English.

- **Cell sheet (or mesh)** — the model of the tissue: a hollow ball of polygonal cells sharing walls, like a football.
- **Operator** — one mechanism as a reusable building block: "cells divide", "chemical spreads", "surface pulls tight". A model is a combination of these.
- **The loop** — the automated cycle: propose a change, predict the outcome, run, measure, record whether the prediction held.
- **Phase** — one of the eight stages above. Our agreed stopping points.

The four measurements, renamed

Agreed 31 July. The old names misled in three separate ways and are being retired: `protr` reads as a length but is a *ratio*; `final` became ambiguous the moment we introduced the evidence horizon; and `Q` said nothing at all. The rename happens as one mechanical pass once the current work lands, with old records aliased on read — the archive is never rewritten.

was	is	means
`protr`	`elongation`	the 95th-percentile cell radius divided by the median. **1.0 is a sphere.** A ratio, not a length — reading 1.62 as an amount of protrusion is the mistake the old name invited.
`protr_peak`	`elongation_peak`	the most elongated it ever got, over *valid* frames only.
`protr_final`	`elongation_at_end`	at the last *valid* frame — not simply the last frame.
`Q`	`elongation_unforced`	what remains after growth and pushing are switched off and the tissue relaxes. **This is the forced-versus-grown test**, and it is the whole reason the campaign exists: we can already make a tube by pushing; the question is whether the tissue will hold one by itself.

My working log is `SESSION_LOG.md`; the findings register is `findings.py` and the premises are `PREMISES.md`. You should not need any of them.